

1. Capability List

#	Capability	Description
1	registerUser / authenticateUser	Register for an account or log in as a Student, Vendor or Administrator.
2	browseMenu	A student looks at food items and prices of a vendor.
3	manageMenu	Food item added, modified or deleted by a vendor and set to available or unavailable.
4	manageCart	A student adds or removes food items before checkout.
5	placeOrder	Central operation: create a paid and confirmed order for a pick-up point.
6	processPayment	Charge the student's chosen payment method for an order.
7	manageOrderLifecycle	Accepts or rejects orders, updates order status or marks order for pickup by vendor.
8	trackOrder	A student enquires about the status of an order.
9	sendNotification	Send a notification to a student or vendor for an order or system update.

Administrator actions (which are the same as above, but are carried out with elevated privileges) are not a distinct capability, but rather the same action above, done with elevated privileges.

2. Service Contracts

Order Service (main service — 3 ops)

Operation	Input	Output	Errors
placeOrder	studentId, cartId, pickupSlotId, paymentMethodId	orderId, status, pickupTime	CART_EMPTY, ITEM_UNAVAILABLE, VENDOR_CLOSED, INVALID_PICKUP_SLOT, PAYMENT_FAILED

Operation	Input	Output	Errors
updateOrderStatus	orderId, vendorId, newStatus	status	ORDER_NOT_FOUND, INVALID_TRANSITION, NOT_OWNER
trackOrder	orderId, studentId	status, statusTimeline	ORDER_NOT_FOUND

Catalogue Service (main service — 3 ops)

Operation	Input	Output	Errors
checkItem	itemId	itemName, price, available	ITEM_NOT_FOUND
listMenu	vendorId	items[] (itemId, name, price, available)	VENDOR_NOT_FOUND
manageMenuItem	vendorId, itemId (optional), name, price, available	itemId, status	VENDOR_NOT_FOUND, INVALID_INPUT, NOT_OWNER

User Service

Operation	Input	Output	Errors
registerUser	name, email, password, role	userId	EMAIL_ALREADY_EXISTS, INVALID_INPUT
authenticateUser	email, password	userId, authToken, role	INVALID_CREDENTIALS
getUserProfile	userId	name, email, role	USER_NOT_FOUND

Payment Service

Operation	Input	Output	Errors
chargePayment	orderId, amount, paymentMethodId	paymentId, status	PAYMENT_DECLINED, INVALID_METHOD

Notification Service

Operation	Input	Output	Errors
sendNotification	userId, message, type	notificationId, status	USER_NOT_FOUND
getNotifications	userId	notifications[] (message, type, createdAt)	USER_NOT_FOUND

Note: No table names, ID formats or SQL anywhere above — each input/output is described in terms of data that will be passed to and/or received by the caller, not how it is stored.

3. Full Specification: placeOrder

Inputs: studentId, cartId, pickupSlotId, paymentMethodId

Outputs: orderId, status (PLACED on success), pickupTime

Error cases:

Error	Trigger
CART_EMPTY	The cart has no items at checkout time.
ITEM_UNAVAILABLE	One or more cart items are out of stock or removed from the menu.
VENDOR_CLOSED	The vendor isn't currently accepting orders.
INVALID_PICKUP_SLOT	The chosen slot doesn't exist, has passed, or is at capacity.
PAYMENT_FAILED	Payment Service declines the charge.

4. Service Property Validation

Service	Reachable	Self-contained	Has a contract	Independent	Loosely coupled
User	Pass	Pass	Pass	Pass	Pass

Service	Reachable	Self-contained	Has a contract	Independent	Loosely coupled
Catalogue	Pass	Pass	Pass	Pass	Pass
Order	Pass	Pass	Pass	Partial	Partial
Payment	Pass	Pass	Pass	Pass	Pass
Notification	Pass	Pass	Pass	Pass	Pass